

Accelerating Solar Energetic Proton Shielding Analysis with a Generalizable Machine Learning Surrogate

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Abstract

Solar energetic particles (SEPs) present a major radiation risk to spacecraft and astronauts because high-energy protons deposit significant energy in biological and electronic components. Existing Monte Carlo simulations primarily model sources other than protons or limited specifically to a set number of materials. Current machine learning and shielding frameworks either are limited by computational costs and/or lack of generalizability across materials for proton sources specifically. This study evaluates different machine learning models to identify a computationally efficient alternative for predicting proton shielding performance.

Shielding effectiveness was assessed through GEANT4 simulations of protons from 10-200 MeV across areal densities of 0-10 g/cm². A dataset of 35 published shielding materials was constructed, with each material characterized using percent composition of C, H, O, N, Si, and density. K-means clustering grouped materials with similar compositions and densities. Synthetic training and testing datasets were generated by blending materials within each cluster and introducing small variations in composition and density.

The XGBoost model performed the best ($R^2 = 0.9984$). The relative errors were 5.4% on the synthetic testing set and 2.9% on published materials, with dose-weighted mean errors of 2.3% and 1.8% respectively. The model required 7.5 μ s/point (batched single-threaded), representing a speedup of approximately 3,847x over GEANT4 simulations. These results demonstrate the potential of machine learning to enable rapid shielding optimization, support hybrid-machine learning through Monte Carlo workflows, and advance scalable materials discovery.